Project Design Phase-I
Solution Architecture

Project Name	Strategic Product Placement Analysis using Tableau
Maximum Marks	2 Marks

Solution Architecture:

The solution architecture for the Strategic Product Placement Analysis project focuses on analyzing retail sales data and generating insights through interactive Tableau dashboards. The architecture connects the dataset, data processing stage, visualization layer, and user interaction components.

The system begins with collecting the dataset containing information about product categories, product placement positions, pricing strategies, promotion status, consumer segments, and sales volume. The dataset is imported into Tableau where the data is processed and prepared for analysis.

In the data processing stage, Tableau organizes the dataset and allows the creation of calculated fields, filters, and parameters. These features enable the analysis of relationships between product placement and sales performance.

The visualization layer consists of multiple charts and dashboards such as bar charts, scatter plots, heatmaps, pie charts, and KPI indicators. These visualizations help users identify trends and patterns in sales data.

Users interact with the dashboards through filters, parameters, and highlight features that allow them to explore data based on product category, promotion status, consumer segment, and traffic level.

Finally, the system generates insights that help retail managers understand which product placement strategies improve sales performance and assist them in making data-driven business decisions.

Example - Solution Architecture Diagram:

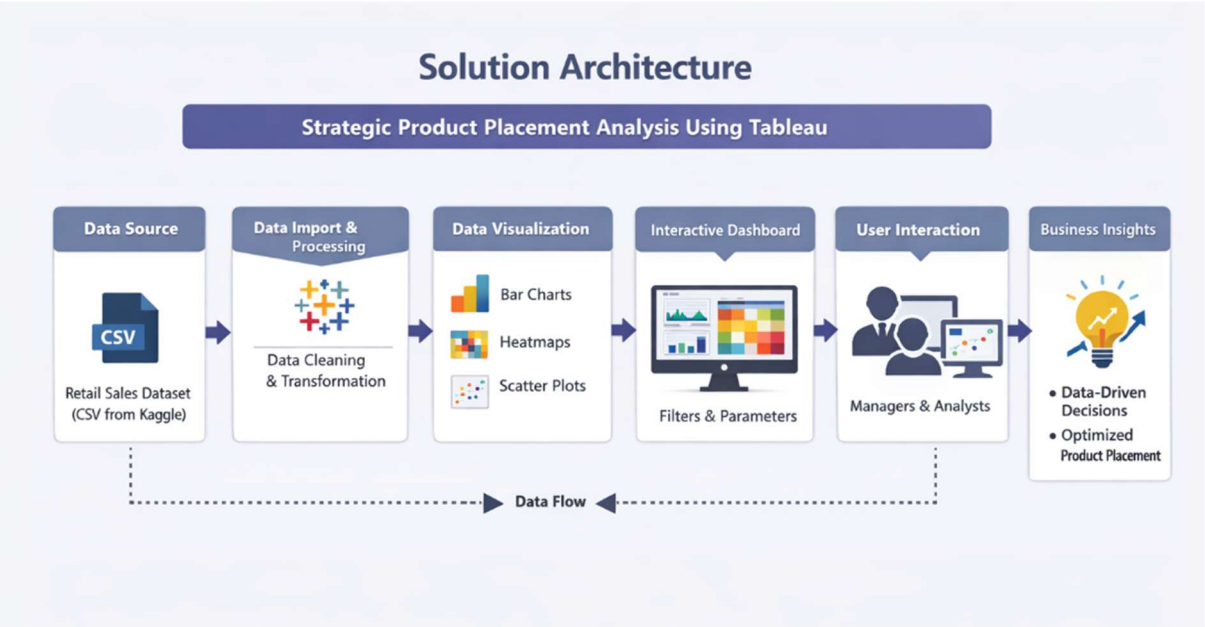

Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/